UTS ANSWER: RESEARCH METHODOLOGY IN COMPUTER SCIENCE

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Number 4: Research Methodology

a. Research Methodology

Our team uses an experimental method with a quantitative approach to test the technical performance of the Shared Fuzzy Logic system.

b. Data Collection

- **Technical Data:**
  Collecting memory metrics and logic execution time through internal system profiling during the simulation.
- **Input Parameter:**
  Health Ratio (H), Euclidean Distance between player and NPC (D), Game stage (S)

c. Experimental Procedure

1. **Fuzzification**
   Converts Numerical Inputs (H, D, S) into linguistic values using membership functions. This step translates raw quantitative data into qualitative categories that the system can interpret more easily.
2. **Inference Engine**
   Applies a shared Fuzzy Inference System (FIS) based on Sugeno rules to determine emotional weights. It evaluates the fuzzy inputs and processes them through a rule base to generate meaningful emotional states.
3. **Defuzzification**

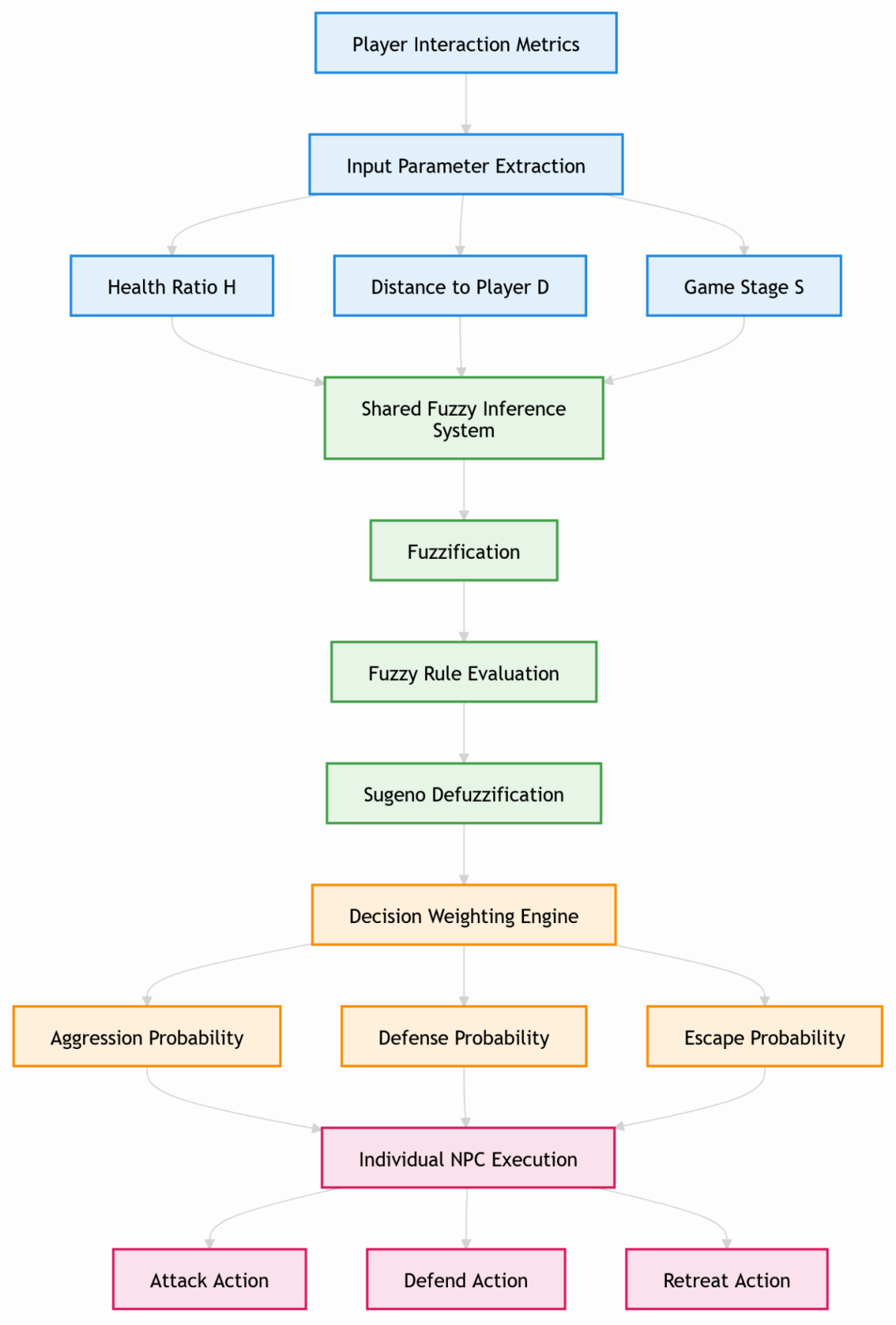
Converts the fuzzy logic output back into concrete, actionable behaviors such as Chase, Patrol, or Retreat. This step ensures that abstract reasoning is translated into executable game actions.

4. **Scalability Test**
   Runs simulations with an increasing number of NPCs to evaluate and validate the efficiency and performance of the shared architecture. This helps determine whether the system can maintain stability under higher computational loads, ensuring it remains effective in large-scale environments.

d. Validity and Reproducibility

- **Validity of Results:**
  - Uses a controlled game prototype so that all AI approaches are tested under identical conditions, ensuring fair results.
  - Employs clear and relevant parameters such as health ratio, Euclidean distance, and game stage that represent realistic gameplay conditions during testing
  - Evaluates performance based on objective metrics such as frame time, memory usage, and CPU storage
  - Conducts behavioral validation to ensure NPCs are not only technically effective, but also exhibit realistic and adaptive behavior.
  - Compares with other methods, such as per-NPC FIS, as well as comparing Mamdani and Sugeno, to confirm performance improvements.
- **Reproducibility:**
  - All tests are conducted repeatedly to obtain stable average results.
  - Uses a well-structured and clearly defined experimental procedure
  - Provides open data and a repository, including game prototype, source code, fuzzy rule base, performance logs, and scripts for generating graphs.

e. System Illustration


Player Interaction Metrics
Input Parameter Extraction
Health Ratio H
Distance to Player D
Game Stage S
Shared Fuzzy Inference System
Fuzzification
Fuzzy Rule Evaluation
Sugeno Defuzzification
Decision Weighting Engine
Aggression Probability
Defense Probability
Escape Probability
Individual NPC Execution
Attack Action
Defend Action
Retreat Action